Purchaser.	No.	Cylinders	Weight	Type	Super-heater	Brick arch	Valve Gear	Mechanical Stoker	Builder
Crowell & Spencer Lumber Co., Ltd.	1	19 x 26	140,000	4-6-0	Yes	No	.....	.....	Baldwin
Cumberland Valley	2	22 x 24	180,300	0-6-0	Yes	Yes	Walschaert	.....	Penn. Alt. shops
Darnell, R. J.	1	.....	64,000	Geared	...	...	.....	.....	Heisler
Detroit & Toledo Shore Line	2	22 x 28	203,000	0-8-0	Yes	Yes	Walschaert	.....	American
Detroit Edison Company	1	21 x 26	140,000	0-6-0	No	Yes	.....	.....	Baldwin
Diamond & Caldor	1	3-11 x 12	120,000	Shay	...	...	.....	.....	Lima
Donora Southern	1	22 x 26	163,000	0-6-0	Yes	Yes	.....	.....	Baldwin
Dubach Lumber Co.	1	3-10 x 12	84,000	Shay	...	...	.....	.....	Lima
East Broad Top	1	20 x 24	161,000	2-8-2	Yes	No	.....	.....	Baldwin
East Tennessee & Western North Carolina	1	16 x 22	100,000	4-6-0	No	No	Walschaert	.....	Baldwin
Elgin, Joliet & Eastern	8	24 x 28	218,000	0-8-0	Yes	Yes	Baker	.....	American
	8	28 x 30	313,900	2-8-2	Yes	Yes	Baker	.....	American
El Paso & Southwestern	5	29 x 30	321,000	2-8-2	Yes	...	.....	Street	American
Elwood Logging Co.	1	3-10 x 10	72,000	Shay	...	...	.....	.....	Lima
Fort Smith & Western	2	22 x 28	200,300	2-8-2	Yes	Yes	Southern	.....	Baldwin
Frost Johnson Lumber Co.	1	3-11 x 12	120,000	Shay	...	...	.....	.....	Lima
Galveston, Harrisburg & San Antonio Ry.	12	27½ x 32	348,000	2-10-2	Yes	No	.....	.....	American
	4	19 x 26	154,000	0-6-0	Yes	No	.....	.....	Baldwin
Garfield Smelting Co.	1	10 x 16	39,000	0-4-0	No	No	.....	.....	Lima
Georgia R. R.	1	27 x 30	280,000	2-8-2	Yes	Yes	.....	.....	Lima
Goodyear Lumber Co.	1	3-12 x 15	150,000	Shay	...	...	.....	.....	Lima
Goodyear Redwood Lumber Co.	1	3-11 x 12	100,000	Shay	...	...	.....	.....	Lima
Grafton & Upton	2	.....	60,000	4-0-4	Electric freight		.....	.....	Gen. Electric
Grand Trunk	10	27 x 30	276,000	2-8-2	Yes	Yes	Baker	.....	Canadian
	5	27 x 30	276,000	2-8-2	Yes	Yes	Young	.....	American
Great Northern	50	28 x 32	229,000	2-8-2	Yes	No	Walschaert	Duplex	Baldwin
	25	26 x 28	220,000	0-8-0	Yes	No	Walschaert	.....	Baldwin
	15	26 x 28	220,000	0-8-0	Yes	No	Walschaert	.....	Baldwin
Green Bay & Western	2	.....	.....	2-6-0	...	...	.....	.....	American
Hammond Lumber Co.	1	3-11 x 12	100,000	Shay	...	...	.....	.....	Lima
Haskell Carpenter Co.	1	3-10 x 12	84,000	Shay	...	...	.....	.....	Lima
Illinois Central	20	26 x 28	278,000	4-6-2	Yes	Yes	Walschaert	.....	American
	20	21 x 26	169,000	0-6-0	Yes	Yes	Walschaert	.....	American
	35	27 x 30	282,700	2-8-2	Yes	Yes	Walschaert	.....	Baldwin
	50	27 x 30	282,700	2-8-2	Yes	Yes	Walschaert	.....	Lima
	25	21 x 26	169,000	0-6-0	Yes	Yes	Walschaert	.....	American
	4	29 x 32	367,000	2-10-2	Yes	Yes	Walschaert	.....	American
Illinois Terminal	1	20 x 24	121,000	2-6-0	No	No	.....	.....	Baldwin
Illinois Traction System	6	.....	120,000	Electric	...	...	.....	.....	Decatur shops
Illinois Zinc Company	1	18 x 24	120,000	0-6-0	No	Yes	.....	.....	Baldwin
Imperial Oil Co., Ltd.	1	2-8 x 12	40,000	Shay	...	...	.....	.....	Lima
Independent Coal & Coke Co.	1	3-8 x 10	56,000	Shay	...	...	.....	.....	Lima
Indiana Harbor Belt	5	23½ x 30	217,000	0-8-0	Yes	Yes	Baker	.....	Lima
Inman Poulson Logging Co.	1	3-11 x 12	100,000	Shay	...	...	.....	.....	Lima
Kanawha, Glen Jean & Eastern	1	22 x 28	218,500	2-8-2	No	No	.....	.....	Baldwin
Kansas City Southern	7	24 x 28	494,000	2-8-8-0	Yes	Yes	Walschaert	.....	American
Kelly Island Lime & Transport Co.	3	3-8 x 8	48,000	Shay	...	...	.....	.....	Lima
Larkins Green Logging Co.	1	.....	106,000	Geared	...	...	.....	.....	Heisler
Lehigh Valley	11	27 x 28	301,500	4-6-2	Yes	Yes	Walschaert	Street	Baldwin
	10	27 x 28	301,500	4-6-2	Yes	Yes	Baker	Street	Baldwin
	30	29 x 32	370,600	2-10-2	Yes	Yes	Walschaert	Street	Baldwin
	10	29 x 32	370,600	2-10-2	Yes	Yes	Baker	Street	Baldwin
Long Pine Lumber Co.	1	3-11 x 12	100,000	Shay	...	...	.....	.....	Lima
Longville Lumber Co.	1	3-12 x 12	130,000	Shay	...	...	.....	.....	Lima
Los Angeles & Salt Lake	6	29½ x 30	357,600	2-10-2	Yes	Yes	Walschaert	.....	Baldwin
Louisville & Nashville	6	23½ x 30	219,000	0-8-0	Yes	Yes	Walschaert	.....	Company shops
	18	27 x 30	302,000	2-8-2	Yes	Yes	Walschaert	.....	Company shops
	18	28 x 30	337,000	2-8-2	Yes	Yes	Walschaert	.....	Company shops
	6	22 x 28	232,000	4-6-2	Yes	Yes	Walschaert	.....	Company shops
Lukens Steel Co.	1	.....	72,000	Geared	...	...	.....	.....	Heisler
Lunkenheimer Co.	1	3-10 x 12	84,000	Shay	...	...	.....	.....	Lima
McKeesport Connecting	2	22 x 28	165,000	0-6-0	No	No	.....	.....	Baldwin
McLean Arkansas Lbr.	1	.....	44,000	Geared	...	...	.....	.....	Heisler
Manufacturers' Ry.	2	.....	160,000	0-4-0	Electric switch		.....	.....	Gen. Electric
Manville Asbestos Co.	2	18 x 24	116,285	0-6-0	...	...	.....	.....	Canadian
Mayer Bros. Construction Co.	2	10 x 16	39,000	0-4-0	...	...	.....	.....	Lima
Michigan Alkali Co.	1	19 x 24	115,000	0-6-0	No	Yes	.....	.....	Baldwin
Michigan Central	10	23½ x 26	276,000	4-6-2	Yes	Yes	Walschaert	.....	American
	10	23½ x 30	217,000	0-8-0	Yes	Yes	Baker	.....	Lima
Middletown Car Co.	1	18 x 24	120,000	0-6-0	No	Yes	.....	.....	Baldwin
Midland Valley	1	22 x 28	186,500	2-8-0	No	Yes	.....	.....	Baldwin
	1	19 x 24	128,000	0-6-0	Yes	Yes	Southern	.....	Baldwin
Milwaukee Coke & Gas Co.	1	.....	.....	0-6-0	...	...	.....	.....	Baldwin
Missouri, Kansas & Texas	1	10 x 16	39,000	0-4-0	No	No	.....	.....	Lima
Mitchell, W. G., Lbr. Co.	1	.....	64,000	Geared	...	...	.....	.....	Heisler
Moekezuma Copper Co.	1	18 x 20	114,000	2-8-0	Yes	No	.....	.....	Baldwin
Monongahela Railway	6	22½ x 30	178,340	2-8-0	Yes	Yes	Baker	.....	American
Monongahela Valley Traction	1	.....	80,000	Electric	...	...	.....	.....	Westinghouse
Morgan & Wright	1	3-12 x 12	120,000	Shay	...	...	.....	.....	Lima
Morgantown & Kingwood	1	22 x 28	181,000	2-8-0	Yes	Yes	Walschaert	.....	Baldwin
Mountain Copper Co.	1	3-10 x 10	72,000	Shay	...	...	.....	.....	Lima
Nashville, Chattanooga & St. Louis	10	25 x 30	272,000	2-8-2	Yes	Yes	Walschaert	.....	Baldwin
Newburgh & South Shore	3	22 x 28	178,000	0-6-0	Yes	Yes	.....	.....	Baldwin
	1	24 x 30	221,700	0-8-0	Yes	Yes	.....	.....	Baldwin
Newport Chemical Works, Inc.	1	18 x 24	100,000	0-6-0	No	Yes	.....	.....	Baldwin
New York Central	35	23½ x 26	272,000	4-6-2	Yes	Yes	Walschaert	.....	American
	45	28 x 28	343,000	4-8-2	Yes	Yes	Walschaert	.....	Lima
	4	23½ x 30	217,000	0-8-0	Yes	Yes	Baker	.....	Lima
New York, Chicago & St. Louis	10	21 x 28	173,000	0-6-0	Yes	Yes	Walschaert	.....	Lima
	5	23½ x 30	217,000	0-8-0	Yes	Yes	Walschaert	.....	Lima
New York, New Haven & Hartford	50	30 x 32	365,000	2-10-2	Yes	Yes	Baker	Duplex	American
	5	.....	360,000	Electric	...	...	.....	.....	Westinghouse
	1	.....	250,000	4-4-4-4	Electric passenger		.....	.....	Gen. Electric
Nimpkish Timber Co.	1	3-12 x 12	120,000	Shay	...	...	.....	.....	Lima
Norfolk & Western	11	24½ & 39 x 32	535,000	2-8-8-2	Yes	Yes	Baker	Duplex	Roanoke shops
	20	23 & 35 x 32	420,000	2-6-6-2	Yes	Yes	Baker	Duplex	American
	20	24½ & 39 x 32	535,000	2-8-8-2	Yes	Yes	Baker	Duplex	Baldwin
North American Sugar Co.	1	.....	.....	2-8-0	...	...	.....	.....	Baldwin
	1	.....	.....	2-6-0	...	...	.....	.....	Baldwin
Northern Ohio Traction & Light Co.	1	.....	100,000	4-0-4	Electric freight		.....	.....	Gen. Electric
Northern Pacific	60	28 x 30	328,000	2-8-2	Yes	Yes	Walschaert	.....	American
	5	26 & 40 x 30	465,000	2-8-8-2	Yes	Yes	Walschaert	.....	American
O'Connell, N. J.	1	3-10 x 12	84,000	Shay	...	...	.....	.....	Lima
Ogden, Logan & Idaho	2	.....	103,000	Electric	...	...	.....	.....	Westinghouse

The Air Brake Association Convention

THE PROCEEDINGS OF THE FIRST DAY'S session of the Air Brake Association convention appeared in last week's issue on page 1245. A brief abstract of the later proceedings follows:

Preparing Air Brakes at Terminals

Installation.—The brake cylinder and auxiliary reservoir should be bolted to the most rigid portion of the underframe and all points of fastening should be equally rigid to avoid twisting or cramping under stresses of brake applications. All air brake piping should be securely clamped to prevent shifting when cars are subjected to shocks. This will avoid many broken or distorted cross-over pipes, loose pipe joints and other causes that produce excessive leakage.

Maintenance.—If the man in charge is to be successful in supervising air brake work, his knowledge should cover the following points: He should know when a triple valve is applied to a car that has been properly repaired, cleaned and tested and is suitable for service. A brake that is tested by making a service application, the reduction of brake pipe pressure being made at the proper rate, will often fail to apply or remain applied, while if it was tested by making the reduction at an excessive rate, which may still be less than required to produce an emergency application, the brake will apply and remain applied. Hence, the importance of testing brakes with a service reduction at the proper rate, this to insure the condition of the brake being such that it will apply and remain applied when placed in the train.

The application of levers of improper dimensions and proportions causes brake-rigging failures, slid flat wheels, improper and unequal braking power and a detrimental influence on train handling. Changing brake shoes without readjusting to between seven and eight inches standing piston travel is often the direct cause of slid flat wheels, break-in-twos and train shocks. Permitting cars to leave shop or repair tracks without hammering and blowing out the brake pipe is not treating the triple valve on the car or other cars in the train in a manner conducive to good operation. Applying triple valves to cars without first seeing that branch pipe strainers are inserted and in good condition and failure to clean out dirt collectors at proper intervals, deprives the triple valves of the protection they are entitled to and causes numerous triple valve troubles.

Inspection Tests.—In making initial inspection tests and retaining valve and brake cylinder tests the committee emphasized the need of giving especial attention to the condition of all piping at points where wear or corrosion is apt to take place. It should be determined that all pipe clamps and nuts on bolts holding cylinders and reservoirs are securely in place. A 90-deg. bent nipple with each leg approximately six inches long, instead of an elbow, should be installed at the triple valve end of the retaining pipe and the pipe should be clamped not closer than six feet from the triple valve to provide flexibility and should be clamped about every six feet from this point to the retaining valve. But one union should be used in the retaining pipe, located at the end of a 90-deg. bent nipple 12 or 14 in. from the triple valve.

Triple valves that do not apply with a service reduction; that apply quick action with a service reduction, and those that have leaks at the exhaust ports should be removed and replaced with others which have been cleaned and tested in accordance with M. C. B practice. Brake cylinders that have leaky packing rings, and the brakes that have not been cleaned for eight months on system cars or 12 months on foreign cars should be cleaned, repaired, tested and stenciled in accordance with M. C. B standard practice.

No train should be permitted to leave a terminal without a terminal test. In testing incoming freight trains at terminals the air brake and the general inspection must not be combined. On both leaving and arriving trains inspectors should rapidly examine the train for piston travel, brakes that fail to apply, brakes that have leaked off and brake pipe leakes. To insure inspectors ascertaining the condition of the brakes on arriving trains, switchmen, carmen and others must not discharge any air from the auxiliary reservoirs or brake pipes of cars that have not been inspected.

Brake Pipe Leakage.—Excessive brake pipe leakage is one of the most liberal contributors to train shocks and break-in-twos. It wastes air, takes away from the engineman the ability to control the amount of brake applications, contributes to brakes sticking, causes overheating of the air compressor and even prevents the maintenance of standard brake pipe pressure. The maximum leakage should not exceed seven pounds per minute, as determined by standard test. This test consists of making a reduction of ten pounds from standard pressure, lapping the brake valve and then noting the leakage during the following minute after the service exhaust ends. By observing the caboose gage during brake applications train men can often note the existence of excessive leakage.

The most common cause for brake pipe leakage is poorly clamped piping that will permit shifting in switch movements, or shocks that occur along the road. Allowing train and yard men to pull hose apart instead of separating them by hand, produces spread couplings jaws, destroys gaskets and creates porous hose.

Brake pipe is not the only leakage to be considered in our effort to eliminate shocks and break-in-twos. Brake cylinder and auxiliary leakage are just as productive of damage to trains as is brake pipe leakage. If a triple valve performs its duty and permits the desired pressure to pass from the auxiliary reservoir to the brake cylinder when the reduction is made and then, due to a bad leather in the cylinder or a defective gasket under the pressure head, the pressure is permitted to leak to the atmosphere, the effectiveness of that brake is either partially or wholly lost. On the other hand, if the brake cylinder and brake pipe are comparatively tight and we have a leaky auxiliary, either from a carelessly applied drain plug, a poorly fitting exhaust valve or a slide valve leaking, then after the application is made the auxiliary pressure leaks down two or three pounds below the brake pipe pressure, the brake will release and its efficiency is lost. Other factors that assist in producing leakage are brake pipes applied out of proper height and distance from the face of the coupler; nipple ends broken off and new threads cut on old nipples, shortening the brake pipe; angle cocks applied and not given the proper angle toward the center of the track.

Defective Triple Valves.—Any number of cases have come under the observation of the writer, when with the brake pipe and auxiliary fully charged the trainman in separating the train would close only the angle cock on the portion of train to be moved, leaving the other one open, and then pull the hose apart, setting all the brakes in emergency on the standing portion of the train, regardless of its length. Repeatedly applying brakes in emergency in this manner is frequently the cause of bent emergency piston stems.

Undesired Quick Action.—Undesired quick action can be caused by permitting triple valves to become dirty and gummy to the extent that the piston sticks to the bushing so that a high differential must be brought about to cause it to move. It can also be caused, especially with long trains, by a very light brake application or a very slow reduction, produced by lapping the brake valve and allow-